SUPPORTING ADULT GUIDANCE

The perpendicular from the centre to a chord

Circle Theorems · Lesson 8. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson establishes that the perpendicular from the centre of a circle to a chord bisects the chord, together with its converse and the equal-chords-equidistant result. Learners discover the bisection by dragging a chord in the chord-bisector lab (the two halves stay equal), prove it by RHS congruence, and then apply it with Pythagoras to find chord lengths and distances from the centre. It is the practical workhorse of the unit — the basis for locating a circle's centre from an arc.

Driving Question: *Why does dropping a perpendicular from the centre always cut a chord in half?*

The mathematics, in plain terms

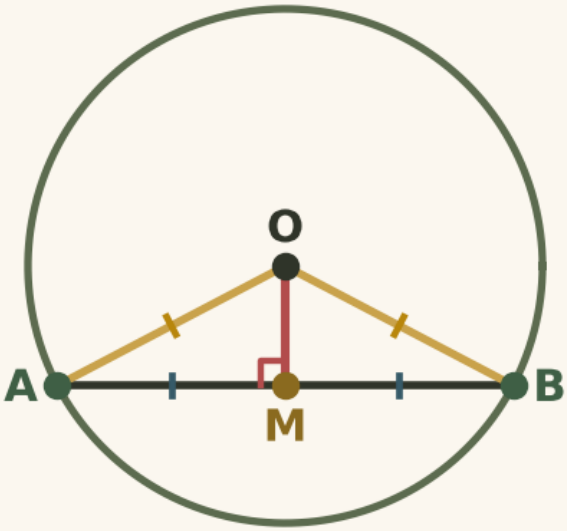
For a chord AB with midpoint M and centre O, OM ■ AB and AM = MB: triangles OMA and OMB have OA = OB (radii), a common side OM, and right angles at M, so they are congruent (RHS). Conversely, if OM meets AB at its midpoint, the same congruence forces the angles at M to be equal and hence 90°. Since OA is a radius, $OA^2 = OM^2 + AM^2$, linking radius, distance and half-chord. Equal chords subtend congruent triangles at the centre and so are equidistant from it.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Find a distance with Pythagoras

A chord AB of length 16 cm is drawn in a circle of radius 10 cm. Find the distance OM from the centre to the chord.

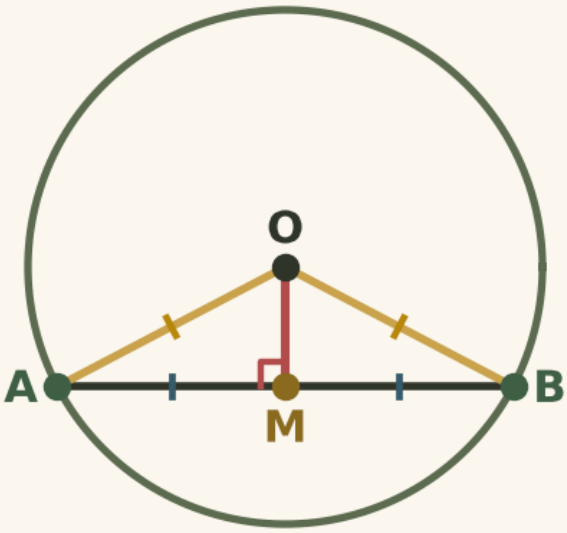


1. The perpendicular from O bisects AB, so AM = 16 ÷ 2 = 8 cm.	perpendicular bisects the chord
2. Triangle OMA is right-angled at M, with hypotenuse OA = 10 (radius).	radius is the hypotenuse
3. Pythagoras: $OM^2 = OA^2 - AM^2 = 10^2 - 8^2 = 36$.	$OM = \sqrt{36}$

OM = 6 cm — the chord is 6 cm from the centre.

Example 2 — Why the perpendicular bisects the chord

Prove that the perpendicular from the centre O to a chord AB bisects it (AM = MB).



1. Drop OM ⊥ AB, so ∠OMA = ∠OMB = 90°.	right angles at M
2. OA = OB (radii, tick marks) and OM is shared.	equal radii, shared side
3. Right angle, hypotenuse OA = OB, one shared side ⇒ congruent (RHS).	triangles OMA ≅ OMB

So AM = MB — the perpendicular from the centre bisects the chord. ■

Discover the halving, prove it, then compute

Let the learner drag a chord and see the two halves stay equal before naming the rule. Prove it cleanly with RHS congruence — the same argument used for the two equal tangents — and state the converse. Only then bring in Pythagoras to find lengths, insisting on a reason at each step and on drawing the radius to expose the right-angled triangle. Introduce equal-chords-equidistant once the core result is secure. This result underpins Lessons 9–10, where several theorems are combined in multi-step problems, and it is how a centre is recovered from an arc.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	See the perpendicular from the centre land on the midpoint.	~9 min
Movement	Use the bisector and Pythagoras to find lengths and distances.	~11 min
Reflection	Chord-bisector lab: drag the chord; the halves stay equal.	~12 min
Creativity	Prove AM = MB by RHS congruence; state the converse.	~9 min
Rest	A short pause on symmetry as a shortcut.	~2 min
Nutrition	Intellectual: how a perpendicular bisector pins down a centre.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Not halving the chord before using Pythagoras.
'The perpendicular bisects the chord — use HALF the chord as the triangle's side, not the whole chord.'

Using the whole chord as a side of the right-angled triangle.
'Which length is opposite the centre? Only half the chord sits in the right-angled triangle.'

Forgetting the radius is the hypotenuse.
'OA is a radius and the longest side — it is the hypotenuse, so radius^2 = distance^2 + half-chord^2.'

Giving a length with no reason.
'Correct — and the reason is "the perpendicular from the centre bisects the chord", then Pythagoras.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: that all radii of a circle are equal, congruent triangles (RHS), and Pythagoras' theorem. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner uses the whole chord in Pythagoras, or gives a length with no reason. The aim is a question that surfaces the halving, the radius, and the right-angled triangle.

- “Where does the perpendicular from the centre meet the chord? So how long is each half?”
- “Which side of your triangle is the radius? Is it the hypotenuse?”

- “Is the length in your triangle the whole chord, or half of it?”
- “What is your reason? Name the fact, then use Pythagoras.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why does the perpendicular from the centre bisect the chord?

Drop the perpendicular OM. Triangles OMA and OMB have equal radii OA = OB, a shared side OM and right angles at M, so they are congruent (RHS); their third sides AM and MB are therefore equal.

Is the converse true?

Yes. If a line from the centre meets a chord at its midpoint, the same congruent triangles force the angles at M to be equal, so each is 90° — the line is perpendicular.

How do I find a chord length or a distance?

Draw the radius to one end of the chord. With the perpendicular, you get a right-angled triangle: $\text{radius}^2 = (\text{distance from centre})^2 + (\text{half-chord})^2$. Solve for the unknown, then double the half-chord if you need the whole chord.

Why are equal chords equidistant from the centre?

Equal chords give congruent triangles at the centre, so their perpendicular distances from the centre are equal — and the further a chord is from the centre, the shorter it is.

Where is this used?

It locates a circle's centre: bisect two chords perpendicularly and the bisectors cross at the centre. This is how you find the centre of a broken plate, an arch or a gear from a fragment of its rim.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.